

The empirical recurrence for OEIS A302805, recovered from its tail

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Abstract

OEIS A302805 records “Empirical recurrence of order 90” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 142$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 93$. Its last coefficient is $c_{90} = 12230590464 \neq 0$, so no further tail satisfies anything shorter; any order-90 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302805 is “Number of nX5 0..1 arrays with every element equal to 0, 1, 2, 3 or 5 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 384, 5117, 74072, 1089773, 15904814, 232260380, 3396923500, ...

The entry states:

Empirical recurrence of order 90 (see link above)

As of the “Last modified” line on the live entry (Apr 13 2018 13:07:10, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 142$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 142$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 90: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 90.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 284$ exact terms from $n = 3$ on, it returns order 90 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{90} c_i a(n-i),$$

c_1	15	c_{24}	423171993026436	c_{47}	8054695485358947595	c_{70}	595286392305132304
c_2	2	c_{25}	564549837034949	c_{48}	2803261924572108820	c_{71}	323546065847184136
c_3	461	c_{26}	636161269178027	c_{49}	7693653032925784757	c_{72}	−387468874371499700
c_4	−8323	c_{27}	−4467217701413588	c_{50}	−11172747752691492448	c_{73}	31543830085642920
c_5	−4577	c_{28}	−1653356870828548	c_{51}	−1758255054307591813	c_{74}	169004713287497192
c_6	−55779	c_{29}	−7273425105592147	c_{52}	−350316139204996250	c_{75}	−107858019846286520
c_7	1660949	c_{30}	33735099607533041	c_{53}	2758105283416588956	c_{76}	32187922420609488
c_8	1183332	c_{31}	−6454396869005823	c_{54}	7604313251962375420	c_{77}	−18181827968566752
c_9	−790765	c_{32}	56449387735165593	c_{55}	−8215124318479492404	c_{78}	4709411051028928
c_{10}	−163149176	c_{33}	−184804645771028845	c_{56}	5572711243437374750	c_{79}	6476709375009664
c_{11}	−100547578	c_{34}	70741904973819128	c_{57}	−10218629893026826047	c_{80}	−7508893256287104
c_{12}	481188647	c_{35}	−284748340957535711	c_{58}	4594805011302016986	c_{81}	4410708546247680
c_{13}	8997275009	c_{36}	737739798111108177	c_{59}	9463281171003090332	c_{82}	−1897235571064320
c_{14}	4218236597	c_{37}	−181388504219838911	c_{60}	−9345225846797996891	c_{83}	654448560039936
c_{15}	−34282216717	c_{38}	855959736885913191	c_{61}	−47258302460454372	c_{84}	−196849836306432
c_{16}	−298152807870	c_{39}	−2105548427959855808	c_{62}	−1294299680941463858	c_{85}	49050398306304
c_{17}	−109380645881	c_{40}	−313498601594815213	c_{63}	4216489712640119082	c_{86}	−7816153853952
c_{18}	1246570455658	c_{41}	−1100269748315139832	c_{64}	−521521515416351626	c_{87}	2324905721856
c_{19}	6126127166514	c_{42}	4074061018812850765	c_{65}	−2275470291171162972	c_{88}	−178303057920
c_{20}	2205272217190	c_{43}	2991731561156685685	c_{66}	892547876495016486	c_{89}	105871048704
c_{21}	−28044989210809	c_{44}	−1510206078265122512	c_{67}	1417462251974223286	c_{90}	12230590464
c_{22}	−77604425771437	c_{45}	−4779026970919336641	c_{68}	−563110655563988574		
c_{23}	−40659024866458	c_{46}	−7417294544680436834	c_{69}	−1019973510548251284		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 93$, and it is the recurrence OEIS A302805 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{90} - c_1x^{90-1} - \dots - c_{90}$. Its constant term is $-c_{90} = -12230590464$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 90 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 90, so $\deg q = 90$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 90, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 90, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 12230590464.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-90 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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